Type MS · Describe virus structure and reproduction

Answer 1 of 3 · Question 9 part b · 6 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

9 (a)	Any three of the following • causes scarring/cirrhosis; • unable to detoxify substances / produce bile; • gene mutation; • leading to cancer;		3
(b)	Any six from the following • virus attaches to (host) cell; • DNA injected into cell; • DNA becomes incorporated into host cell DNA; • viral DNA replicated along with host cell DNA; • protein synthesis; • new virus produced; • lysis of host cell; • new viruses released (to infect new cells);		6
(c)	• antigens injected; • white cells/lymphocytes produce antibodies; • memory cells produce antibodies faster;		3
			Total 12

Type MS · Describe virus structure and reproduction

Answer 1 of 3 · Question 9 part b · 6 marks

2022 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

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Type MS · Describe virus structure and reproduction

Answer 2 of 3 · Question 2 part a.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
2	(a)	(i)	• genetic material made of RNA (1) • no DNA (1)		2
		(ii)	<table><tr><th>Disease</th><th>Blood Tested (✓)</th></tr><tr><td>anaemia</td><td></td></tr><tr><td>cystic fibrosis</td><td></td></tr><tr><td>gonorrhoea</td><td>✓</td></tr><tr><td>HIV</td><td>✓</td></tr><tr><td>scurvy</td><td></td></tr></table>	-1 for each extra tick	2
		(iii)	• donor/blood transfused into person with Ebola (1) • needs to be compatible/matched/same group/not rejected (1) • otherwise agglutination/clumping occurs (1)	R clotting	3
		(iv)	• cause blood cells to burst/cells are damaged/destroyed/lose structure (1) • results in loss of function (1)		2
		(v)	• antibodies in donated blood (1) • can destroy virus in infected person (1)		2
	(b)		• uracil present, not thymine (1) • guanine would pair with cytosine/adenine with uracil (1) • percentage of G and C/A and U would be the same (1)		3

Total for question = 14 marks

Type MS · Describe virus structure and reproduction

Answer 2 of 3 · Question 2 part a.i · 2 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Describe virus structure and reproduction

Answer 3 of 3 · Question 8 part a.i · 2 marks

2023 JUN · Paper 02 (regional R) · find part a.i in this mark scheme extract

8	(a) (i)	Any two from • DNA/RNA/genetic material; • surrounded by a protein coat/capsid; • (protein) spikes on outer surface;		2
	(ii)	Any five from • transcription (of DNA); • into mRNA; • translation (of mRNA) at the ribosome; • by tRNA; • reference to (tRNA) anticodon pairing with (mRNA) codon; • amino acids assembled to form (viral) proteins;		5
	(b) (i)	Human Immunodeficiency virus/HIV;		1
	(ii)	• (1400000 – 650000) = 750000; • (750000 ÷ 1400000) (x 100) = 54%	Allow any figure that rounds to 54	2

Total for Question 8 = 10 Marks